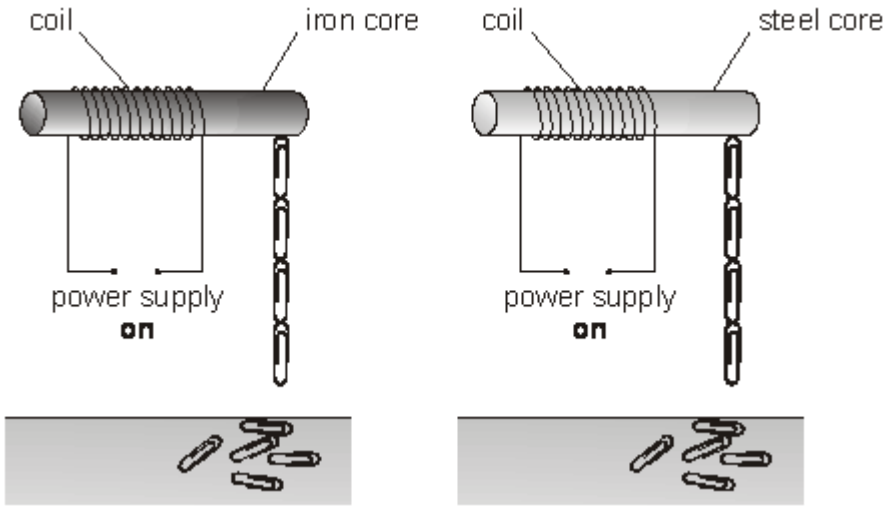
Q1. David made two electromagnets as shown below.
He used paper-clips to test the strength of each electromagnet.
He switched on the power supply in both circuits.



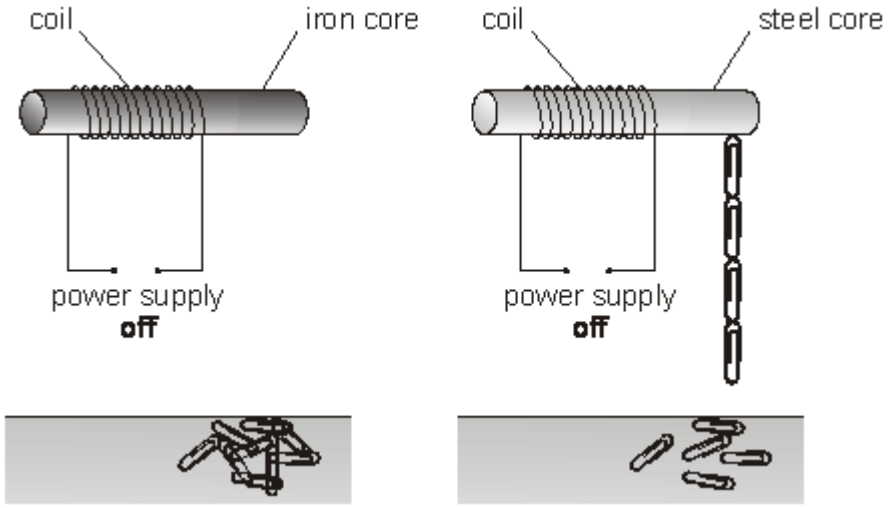
(a) How can you tell that the strength of both electromagnets is the same?

.....

.....

1 mark

(b) David switched off the power supply in both circuits.
The paper-clips fell off the iron core, but **not** off the steel core.



Why is iron used, rather than steel, for the core of an electromagnet?
Use the diagrams above to help you.

.....

.....

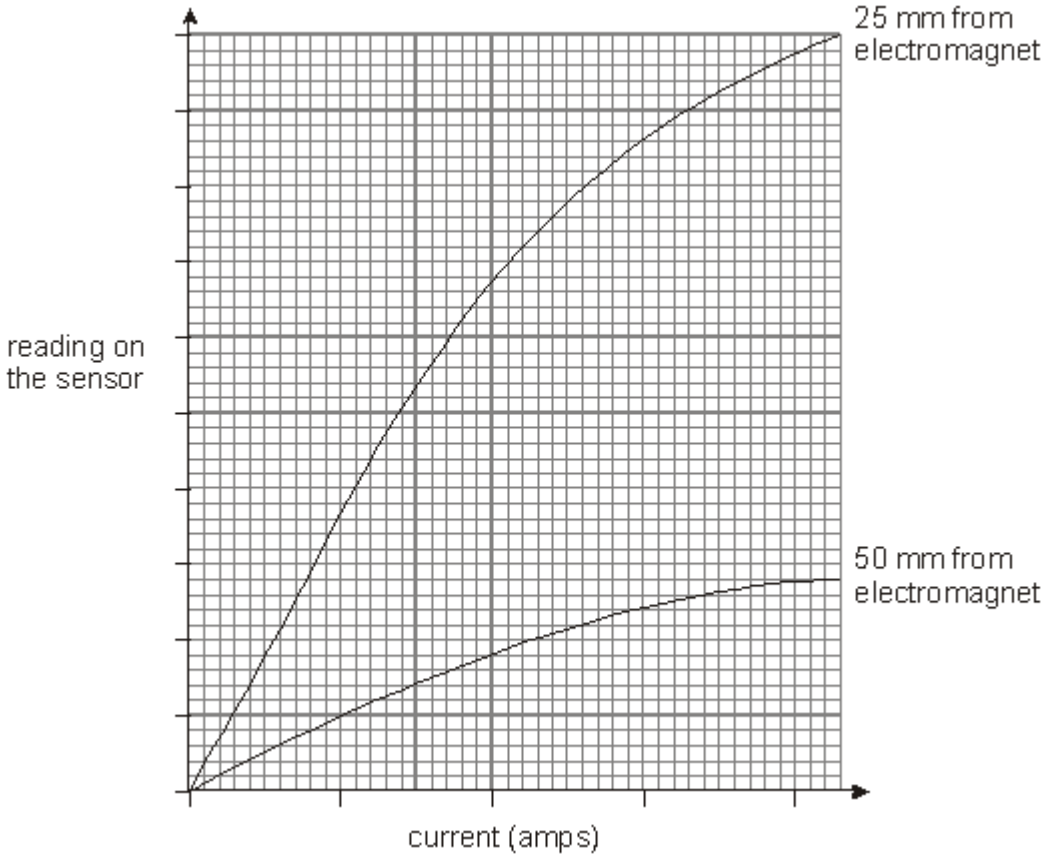
1 mark

(c) David used a sensor to measure the strength of an electromagnet.

He placed the sensor 25 mm from the electromagnet and increased the current in the coil.

He repeated the experiment with the sensor 50 mm from the electromagnet.

The graph below shows his results.



(i) How did the distance of the sensor from the electromagnet affect the reading on the sensor?

.....

.....

1 mark

(ii) How did the size of the current in the coil affect the strength of the electromagnet?

.....

.....

1 mark

(iii) What else could David do to an electromagnet to change its strength?

.....

.....

1 mark

maximum 5 marks

Mark schemes

Q1.

(a) • both picked up the same number **or** four paper-clips

accept 'they both picked up the same number'
accept 'same amount of paper-clips'
accept 'there were 5 out of 9 paper-clips left for both'
accept 'the same mass of paper-clips'
'they hold the same clips' is insufficient

1 (L5)

(b) any **one** from

- it does not stay magnetised
- it can be turned off

accept 'you cannot turn steel off'

- objects do not stay attached to it
- iron loses its magnetism
- steel stays magnetised

1 (L6)

(c) (i) any **one** from

- the greater the distance the lower the reading
- the further away the smaller the reading

accept the converse
accept 'at big distance the field is weaker' or the converse
accept 'at 50 mm the reading is lower'
accept the converse
*do **not** accept 'the bigger the distance the smaller the amps **or** current'*

1 (L6)

(ii) • the greater the current the stronger the electromagnet

1 (L6)

(iii) any **one** from

- change the number of turns
- change the thickness of the wire
- change the diameter of the core

accept 'use more coils'
accept 'use fewer or less coils'
*accept 'put the coils closer together' **or** the converse*
accept 'change the metal of the coils'
accept 'use a different sized core'
*accept 'use nickel **or** cobalt core'*

accept 'use a different core'
'use bigger coils' is insufficient
'use more wire' is insufficient
*do **not** accept 'add more batteries'*

1 (L6)

[5]